

SANMATHI PARAMESWARAN

M.Sc. Water Resources Engineering student and Erasmus research intern

✉ sanmathiparames@gmail.com ☎ +4917634461557 📍 4A, Heilmannstrasse, Stuttgart-70190

🌐 www.linkedin.com/in/sanmathi-parameswaran-ab1318299

🔗 Kasim0616.github.io/sanmathi-portfolio

SUMMARY

Proactive MSc Water Resources Engineering student with a high degree of independence in managing complex research projects, water quality analysis, and sustainability assessments. Currently spearheading a Master's thesis on wastewater-based green hydrogen production in a fast-paced research environment.

EXPERIENCE

Drees & Sommer / University of Stuttgart, Research Thesis

03/2026 – Present

Remote

- Independently managing the end-to-end research lifecycle for green hydrogen production using advanced treated wastewater.
- Designing high-purity water treatment systems, including Ultrafiltration (UF), Reverse Osmosis (RO), and Electrodeionization (EDI)
- Developing sustainable pathways through Techno-economic modeling to evaluate CO2 reduction and cost efficiency
- Proactively optimizing process efficiency and conducting Life Cycle Assessments (LCA) for membrane filtration technologies.
- **Technical Skills:** Green Hydrogen Technology, LCA, UF/RO/EDI, Techno-economic Modeling, Membrane Filtration.

Norwegian University of Science and Technology (NTNU),

12/2025 – 03/2026

Erasmus Research Intern

Trondheim, Norway

- Analyzed applied research on stormwater pollutant clearance and multi-barrier water quality monitoring.
- Processed large environmental datasets to model and behavior of persistent mobile organic compounds, specifically PFAS, PAH, Heavy metals in urban runoff.
- Demonstrated a high level of drive and independence while working in an international research setting, specifically in automating data evaluation workflows using Python and RDKit.
- Explicitly state that you collaborated with an interdisciplinary team of chemists and data scientists to automate workflows
- **Technical Skills:** Python, RDKit, EPI Suite, Environmental Data Science, Pollutant Modeling, Research Automation.

AquaAware, Scientific Assistant

05/2025 – 11/2025

Stuttgart, Germany

Onsite

- Conducted laboratory water quality testing and experimental validation for a mobile monitoring application.
- Performed sample preparation for HPLC and GC-MS analysis to detect trace-level contaminants.
- Independently executed laboratory testing protocols and authored technical documentation for real-time water sensing products with minimal supervision.

- Supported applied R&D activities and early-stage product development for real-time water sensing.
- **Technical Skills:** Lab Analysis (HPLC, GC-MS), Micropollutant Assessment, Technical Writing, Experimental R&D.

Srusti Traders and Builders, Environmental Trainee Engineer	01/2021 – 06/2021
• Drafted detailed technical site plans and utility layouts for environmental infrastructure using AutoCAD (2D/3D). • Developed 3D models for drainage systems and sustainable construction projects in Revit Architecture. • Performed on-site collection and analysis of water and soil samples to ensure regulatory compliance. • Compiled field data into technical reports, improving documentation and site monitoring efficiency. • Technical Skills: AutoCAD (2D/3D), Revit Architecture, BIM, Site Sampling (Water/Soil), Environmental Compliance.	Tirunelveli, India Onsite

EDUCATION

University of Stuttgart, Stuttgart, Germany, <i>M.Sc. Water Resources Engineering and Management (WAREM)</i>	10/2023 – Present
National Engineering College, Tamil Nadu, India, <i>Bachelor of Civil Engineering</i>	08/2018 – 06/2022

LANGUAGES

Tamil – Mother Tongue	English – C1	German – B1	Hindi – A2
Norwegian – A1			